

## **Comparison of Regions Based on Sales**

### **Course-end Project 1**

#### **Project Description**

This project aims to compare the sales performance of the two selected regions in the Sample Superstore database. The organization would like to learn which region performs well and which region requires improvement. Comparing sales data for different metrics, such as total sales, average sales per order, number of customers, number of orders, number of products, and date of first order is done through a Tableau dashboard.

#### **Objective**

The primary objective of this assignment is to develop a Tableau dashboard that enables management to compare sales for two regions chosen by them.

**Datasets:** Sample Superstore

#### **Steps Performed in Tableau**

##### **1. Select Dataset**

The **Sample Superstore** dataset was selected and loaded into Tableau.

##### **2. Create Folder**

After loading the Sample Superstore dataset into Tableau, the Orders sheet was selected for analysis. The required data fields were organized properly to make the dashboard development process easier.

The Group by Folder option was used in the Data Pane to organize important fields such as Customer Name and Order ID. This helped separate customer and order-related information and made the dataset easier to manage.

A folder was created for customer and order details, including fields such as:

Customer Name

Customer ID

Order ID

### **3. Location Hierarchy**

A hierarchy named Location was created using the Country field.

The Location hierarchy includes:

Country

State

City

This hierarchy allows users to drill down from country level to state level and then city level. It makes geographical analysis easier and improves the dashboard's usability.

### **4. Parameters Created**

Two parameters were created to allow dynamic comparison between two selected regions.

The parameters are:

Primary Region

Secondary Region

Both parameters include all available regions from the dataset:

Central

East

South

West

These parameters allow users to select any two regions and compare their performance in the dashboard.

### **5. Calculated Fields**

Several calculated fields were created to compare the selected regions and display the required dashboard values.

#### **Primary Region Sales**

This calculated field was created to show sales only for the selected Primary Region.

Formula:

IF [Region] = [Primary Region] THEN [Sales] END

#### **Secondary Region Sales**

This calculated field was created to show sales only for the selected Secondary Region.

Formula:

IF [Region] = [Secondary Region] THEN [Sales] END

### **Selected Regions**

This calculated field was created to filter the dashboard and display only the selected Primary Region and Secondary Region.

Formula:

[Region] = [Primary Region] OR [Region] = [Secondary Region]

### **Region Type**

This calculated field was created to identify whether the selected region is the Primary Region or Secondary Region.

Formula:

IF [Region] = [Primary Region] THEN "Primary Region"  
ELSEIF [Region] = [Secondary Region] THEN "Secondary Region"  
END

## **6. First Order Date**

This calculated field was created to identify the first order date for each region.

Formula:

{ FIXED [Region] : MIN([Order Date]) }

### **Key Performance Indicators**

The dashboard displays the following KPIs for both the Primary Region and the Secondary Region:

First Order Date

Total Sales

Average Sales per Order

Number of Customers

Number of Orders

Number of Products in Sale

### **Total Sales**

Total Sales shows the overall sales amount generated by the selected region.

Formula:

`SUM([Sales])`

### **Average Sales per Order**

Average Sales per Order shows the average sales value for each order.

Formula:

`SUM([Sales]) / COUNTD([Order ID])`

### **Number of Customers**

Number of Customers shows how many unique customers purchased from the selected region.

Formula:

`COUNTD([Customer Name])`

### **Number of Orders**

Number of Orders shows the total number of unique orders placed in the selected region.

Formula:

`COUNTD([Order ID])`

### **Number of Products in Sale**

Number of Products in Sale shows the number of unique products sold in the selected region.

Formula:

`COUNTD([Product Name])`

### **First Order Date**

First Order Date shows the earliest order date available for the selected region.

Formula:

`MIN([Order Date])`

## 7. Dashboard Creation

The dashboard is titled Sales Comparisons by Region. It is divided into two main sections. The left side shows the Primary Region, which is Central. The right side shows the Secondary Region, which is South.

Each side of the dashboard contains the same type of information, which makes comparison easy. The dashboard includes KPI cards, regional maps, sub-category tables, line graphs, and bar graphs.

The dashboard includes:

Primary Region parameter

Secondary Region parameter

First Order Date

Total Sales

Average Sales per Order

Number of Customers

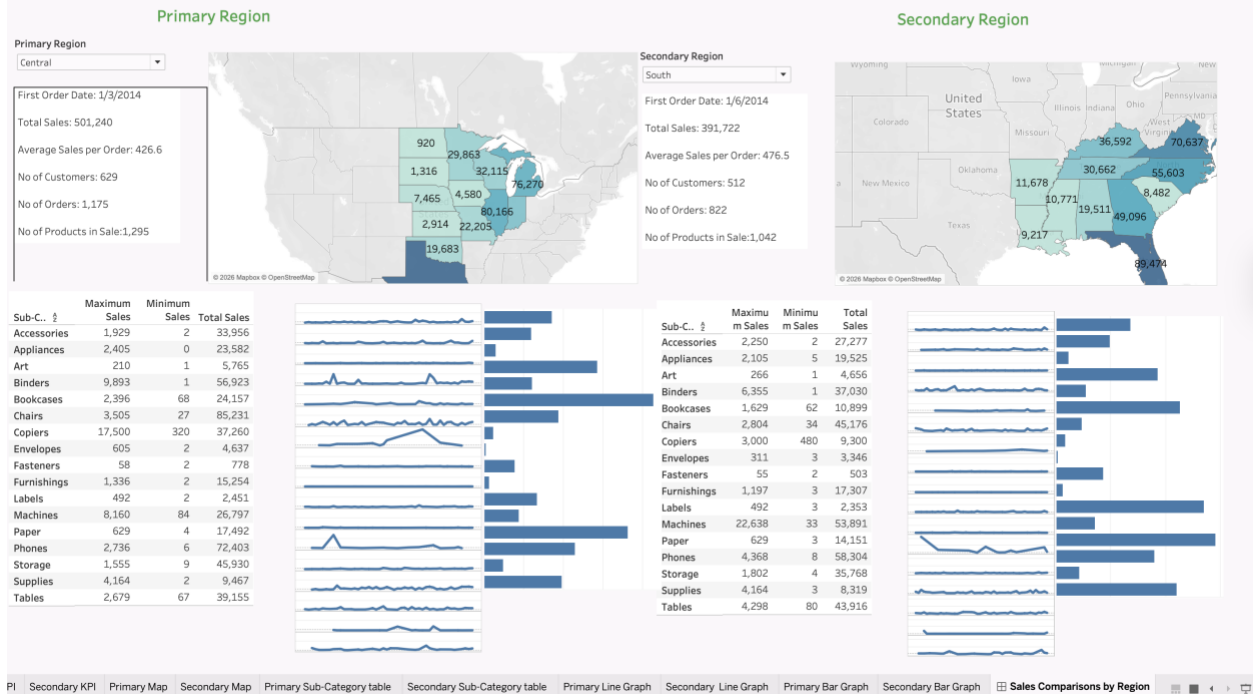
Number of Orders

Number of Products in Sale

Line graph

Bar graph

## Sales Comparisons by Region



### Primary Region Performance: Central

The primary region that is selected in the dashboard is Central.

In Central region, the first order date is 1/3/2014. The total sales is 501,240. The average sales per order is 426.6. The number of customers is 629, number of orders is 1,175, and number of products in sale is 1,295. Thus, it is clear that Central region has high number of customers and orders. Moreover, Central region has also high number of products in sale, implying wide product reach for Central region.

### Secondary Region Performance: South

The secondary region that is selected in the dashboard is South.

In South region, the first order date is 1/6/2014. The total sales is 391,722. The average sales per order is 476.5. The number of customers is 512, number of orders is 822, and number of products in sale is 1,042. Thus, it is evident that South region has less number of customers and less number of orders compared to Central region. However, the average sales per order of South region is higher compared to Central region.

## **KPI Comparison**

From the KPI cards, the Central region outperforms the South region in terms of total sales, customers, orders and number of products in sale. Total sales for the Central region stands at 501,240 whereas that of South region is 391,722. It means that the Central region registered higher total sales. The number of customers of the Central region is 629 whereas that of South region is 512. It means that the Central region has more customers. The number of orders for the Central region is 1,175 whereas for the South region is 822. It means that Central region had higher orders. The number of products in sale in Central region is 1,295 whereas that of South region is 1,042. It means that Central region had more sales. On the other hand, the average sales per order in South region is higher than that in Central region. The average sales per order in South region is 476.5 while in Central is 426.6.

## **Map Visualization Explanation**

Map Visualization presents sales by state in selected regions.

In case of the Central region, the map provides information on sales by states like Texas, Illinois, Michigan, Indiana, Wisconsin, Minnesota, and surrounding states. There are some states in the Central region that have high sales compared to others. Thus, the map will be helpful in determining which states contribute significantly to overall regional sales.

In the South region, the map presents sales by states like Florida, North Carolina, Virginia, Georgia, Tennessee, Kentucky, Louisiana, Mississippi, and Arkansas. There is a high sales value by Florida as compared to most other states in the South region.

Map visualization is helpful in giving a geographical perspective to sales.

## **Sub-Category Table Explanation**

Sub-category wise sales performance is captured in the sub-category table below. The table provides maximum sales, minimum sales, and total sales for each sub-category of products.

Some of the high performing sub-categories in the Central region include Chairs, Phones, Binders, Storage, and Tables. The Chairs sub-category leads in total sales in the Central region, while Phones and Binders follow.

On the other hand, some of the high performing sub-categories in the South region include Phones, Paper, Bookcases, Tables, and Binders. Phones sub-category leads in total sales in the South region.

Table is useful to the management in knowing the high selling sub-categories in each region.

## **Line Graph Explanation**

The line graphs show sales trends over time for each product sub-category. Each small line graph represents how sales changed over the selected time period.

In the Central region, some sub-categories show more fluctuation, meaning sales increased and decreased at different times. Sub-categories such as Binders, Chairs, Phones, and Storage show noticeable changes over time. These changes may indicate seasonal demand or occasional high-value orders.

In the South region, the line graphs also show sales movement over time. Some sub-categories such as Phones, Paper, Tables, and Bookcases show stronger activity compared with other categories. The line graph helps show whether sales are stable, increasing, decreasing, or irregular.

Overall, the line graphs help management understand sales patterns over time. Peaks represent strong sales periods, while flat or low lines show weaker sales periods.

## **Bar Graph Explanation**

The bar graphs compare total sales by sub-category for each selected region. Longer bars represent higher total sales, while shorter bars represent lower total sales.

In the Central region, the longest bars are for Chairs, Phones, Binders, Storage, and Tables. This shows that these sub-categories are the strongest contributors to Central region sales. Chairs appear to be the leading sub-category in the Central region.

In the South region, the longest bars are for Phones, Paper, Bookcases, Tables, and Binders. This shows that these sub-categories are the strongest contributors to South region sales. Phones appear to be the leading sub-category in the South region.

The bar graphs make it easy to compare product performance visually. Management can quickly see which product sub-categories are performing well and which ones are weaker.

## **Dashboard Insights**

The dashboard demonstrates that Central region outperforms the South region in terms of total sales, total number of customers, total number of orders, and total number of products on sale.

Central total sales are 501,240, while South total sales are 391,722, which indicates better sales performance in the Central region. Central region has a larger customer base and a larger number of orders, which means it has a bigger market and higher customer activity.

In spite of having lower total sales and total orders, the South region has a higher average sales per order, which indicates higher expenses per order in the South region. As far as product



performance is concerned, chairs have excellent performance in the Central region, while Phones are the leading sub-category in the South region.

### **Suggested Improvements**

The South region should focus on increasing its number of customers and orders because it has lower total sales than the Central region. Since South already has a higher average sales per order, increasing the number of orders could significantly improve total sales.

The organization can improve South region performance by increasing marketing campaigns, offering promotions, improving product availability, and focusing on strong sub-categories such as Phones, Paper, Bookcases, and Tables.

For the Central region, the organization should continue supporting strong categories such as Chairs, Phones, Binders, Storage, and Tables. The company should also maintain customer engagement because Central has a larger customer base and higher order volume.

### **Conclusion**

In conclusion, the Tableau dashboard effectively compares the two regions, Central and South, based on the data from the Sample Superstore. It reveals that the Central region has larger total sales, more customers, orders, and products on sale than the South region, thus being a dominant one.

On the other hand, despite having lower total sales, less number of customers, and lower number of orders, the South region demonstrates larger average sales per order. Thus, there is room for improving the total number of customers and orders in South. The use of line graphs helps to analyze the sales dynamics while the use of bar charts makes it possible to compare the total sales by sub-category.